Saad Ghani, Ph.D. Student.

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Education

2022 – · · · · **Ph.D. Computer Science, George Mason University.**
Research Focus: Self-Supervised Learning for Motion Planning in Dynamic Environments.

2016 – 2020 **B.Eng. Computer Science, The University of Hong Kong**.
First Class Honours.

2014 – 2016 **A-Levels, Karachi Grammar School, Karachi, Pakistan**.
Further Mathematics (A), Mathematics (A), Physics (A), Chemistry (A).

Employment History

2022 – · · · · **Graduate Teaching Assistant.** Dept. of Computer Science, George Mason University.

2023 – 2025 Summers **Graduate Research Assistant.** RobotiXX Lab, George Mason University.
Supervisor: Prof. Xuesu Xiao

2021 – 2022 **Research Assistant.** Vision Lab, The University of Hong Kong.
Supervisor: Prof. Kenneth K.Y. Wong

2020 – 2021 **Teaching Assistant.** Dept. of Computer Science, The University of Hong Kong.

2018 – 2019 **Undergraduate Research Assistant.** Dept. of Mathematics, The University of Hong Kong.

Research Projects

2024 – · · · · **LfH-CP**, RobotiXX Lab, George Mason University.
Self-supervised motion planner trained to navigate dynamic environments by factorizing hallucination into two stages to generate richer, more diverse obstacle trajectories. [Paper]

2025 **Failing Gracefully**, RobotiXX Lab, George Mason University.
Novel safety formulation that minimizes the impact of failing.
Developed FailBench, a MuJoCo-based simulation framework that injects failures to study robot-environment interactions.
Accepted at ICRA 2026. [Paper]

2023 – 2024 **Dyna-LfLH**, RobotiXX Lab, George Mason University.
Self-supervised method to generate dynamic obstacles to train motion planners to navigate in dynamic environments.
Presented at ICRA 2025 SRL Workshop. [Poster]
Published at IROS 2025. [Paper] [Video] [Poster]

2021 – 2022 **Medical Imaging & Clinical Prediction**, Vision Lab, The University of Hong Kong.
Developed deep learning pipelines for three clinical tasks: knee osteoarthritis severity grading from X-rays (with multi-model benchmarking), 3D echocardiogram landmark prediction, and patient management plan prediction.

2019 – 2020 **Early Detection of Red Tides**. Research Internship & Final Year Project, The University of Hong Kong.
Image classification and Time-Series analysis of Red Tide causative algal species.

Awards

2026 **Outstanding Graduate Teaching Assistant**, George Mason University.

2025 **Summer Research Award**, Institute for Digital InnovAtion, George Mason University.

Awards (continued)

2024 **Digital Innovation Summer Experiential Fellowship**, Institute for Digital InnovAtion, George Mason University.

2023 **Summer Research Award**, Dept. of Computer Science, George Mason University.

2017 **Dean's Honours List**, The University of Hong Kong.

2016-2020 **HKSAR Government Scholarship Fund**, The University of Hong Kong.

Research Publications

1. D. M. Nguyen, **S. A. Ghani**, A. Marshall, A. Andreyev, G. J. Stein, and X. Xiao, "Failing gracefully: Mitigating impact of inevitable robot failures," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
2. X. Xiao, Z. Xu, **S. A. Ghani**, A. Datar, D. Song, P. Stone, A. Mazen, K. Yazdipaz, I. Mateyaunga, M. Faied, et al., "Autonomous ground navigation in highly constrained spaces," *IEEE ROBOTICS & AUTOMATION MAGAZINE*, vol. 186, pp. 1070–9932, 2026.
3. **S. A. Ghani**, Z. Wang, P. Stone, and X. Xiao, "Dyna-lflh: Learning agile navigation in dynamic environments from learned hallucination," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 17 719–17 724. DOI: 10.1109/IROS60139.2025.11247157.

Teaching Experience

George Mason University

Guest Lecture — Delivered guest lectures on Recursion to two undergraduate sections in Object-Oriented Programming, receiving strong positive feedback in anonymous post-lecture surveys.

Courses TA'd — Object-Oriented Programming, Data Structures.

The University of Hong Kong

Courses TA'd — Java and Object-Oriented Programming, Computer Programming I & II, Design & Analysis of Algorithms, Computer Networking, Softwawre Engineering, Computer Organization.

Professional Service

2025 – 2026 **Competition Organizer**, The Benchmark Autonomous Robot Navigation (BARN) Challenge, ICRA.

2024 – · · · · **Journal Reviewer**, IEEE Robotics and Automation Letter.

Conference Reviewer, IEEE International Conference on Robotics and Automation (ICRA).

Conference Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Technical Skills

Coding — Python, Java, C++.

Frameworks & Libraries — ROS, ROS2, PyTorch, MuJoCo, Gazebo, Plotly.

Robots — Clearpath Jackal, Franka Panda.

Languages — English, Urdu.